Monthly High Timing Patterns in NIFTY 50 Stocks: Evidence of Late-Month Bias

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Running title *– Late-Month Bias in NIFTY 50 Highs*

Abstract

This study examines the timing of monthly price highs in NIFTY 50 constituent stocks over a ten-year period (2015–2025), building on earlier evidence of calendar anomalies and seasonality in equities (as documented in prior studies of monthly and seasonal effects). Each trading month was divided into three calendar clusters: Early (1–10), Mid (11–20), and Late (21–31). Analysis of 5,781 monthly observations reveals a statistically significant concentration of monthly highs in the Late cluster ($\chi^2 = 400.6$, $p < 2.2 \times 10^{-16}$), consistent with behavioral trading cycles and liquidity-driven end-of-month effects described in earlier market microstructure research. Sector-level tests ($\chi^2 = 33.13$, $p = 0.51$) indicate that this late-month bias persists across industries, highlighting it as a broad market phenomenon rather than a sector-specific anomaly. These findings contribute to the growing literature on behavioral finance and market timing, offering insights into the persistence of timing-based patterns in Indian equities and their potential applications in tactical portfolio design, short-term trading, and risk management.

Keywords: calendar anomalies, NIFTY 50, monthly price highs, behavioral finance, Indian equity markets

Introduction

Calendar effects have long been a focus of empirical finance research, revealing that asset returns often exhibit predictable patterns tied to specific points in the trading calendar. Classic studies document the turn-of-the-month effect (returns abnormally high around month-ends and beginnings; Ariel, 1987; Lakonishok & Smidt, 1988) and the day-of-the-week effect (Cross, 1973; French, 1980), suggesting that investor behavior and institutional trading flows may drive systematic timing anomalies.

While many of these effects have been examined extensively in developed markets, evidence from emerging economies like India remains limited, particularly with respect to the timing of monthly price peaks. Prior research has focused predominantly on return anomalies rather than the occurrence of intra-month high prices, leaving a gap in understanding whether stock prices tend to make their monthly highs in predictable calendar clusters.

The motivation for this study lies in the potential behavioral and structural drivers behind such clustering. Institutional trading schedules, portfolio rebalancing near month-end, and settlement cycles could influence the timing of price highs. Furthermore, understanding these patterns may offer practical insights for portfolio managers and market participants in optimizing trade timing and managing execution risk.

This paper seeks to address two key questions:

1. Do monthly stock price highs cluster within specific parts of the month?
2. Does this clustering vary across industry sectors, or is it a broader market phenomenon?

By analyzing a decade of data on NIFTY 50 stocks (2015–2025), we aim to contribute to the literature on calendar anomalies by documenting systematic patterns in the timing of monthly highs within the Indian equity market.

Data and Methodology

This study uses daily OHLC (Open–High–Low–Close) price data for all NIFTY 50 constituent stocks covering the period January 2015 to July 2025. Data were sourced from Yahoo Finance using the `quantmod` package in R. For each stock and month, we identified the date corresponding to the maximum closing price, yielding a total of 5,781 stock-month observations.

To investigate the timing of monthly price peaks, each month was divided into three calendar clusters:

- Early Month: Days 1–10
- Mid Month: Days 11–20
- Late Month: Days 21–31

The primary analysis examined whether monthly highs were uniformly distributed across these clusters. We performed a Chi-squared goodness-of-fit test, comparing the observed frequencies of monthly highs in each cluster against the null hypothesis of equal distribution.

To explore whether the timing of monthly highs varied across industry sectors, we conducted a Chi-squared test of independence using a contingency table of sector classifications versus monthly-high clusters. Sector data was based on NSE's official classification of NIFTY 50 companies.

All statistical analyses were performed in R (version 4.3.0), and graphs were generated using ggplot2.

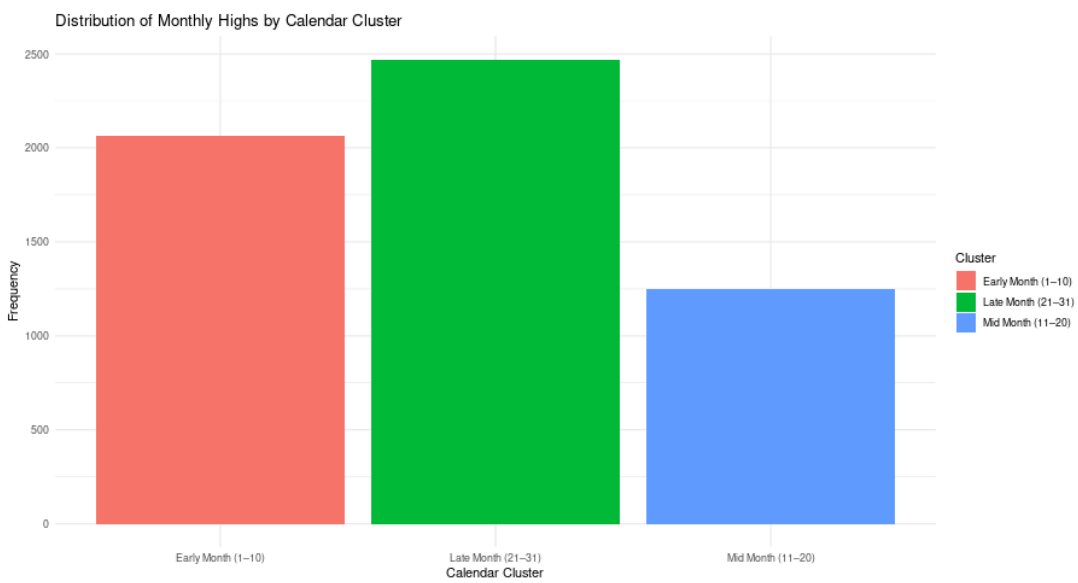

Figure 1

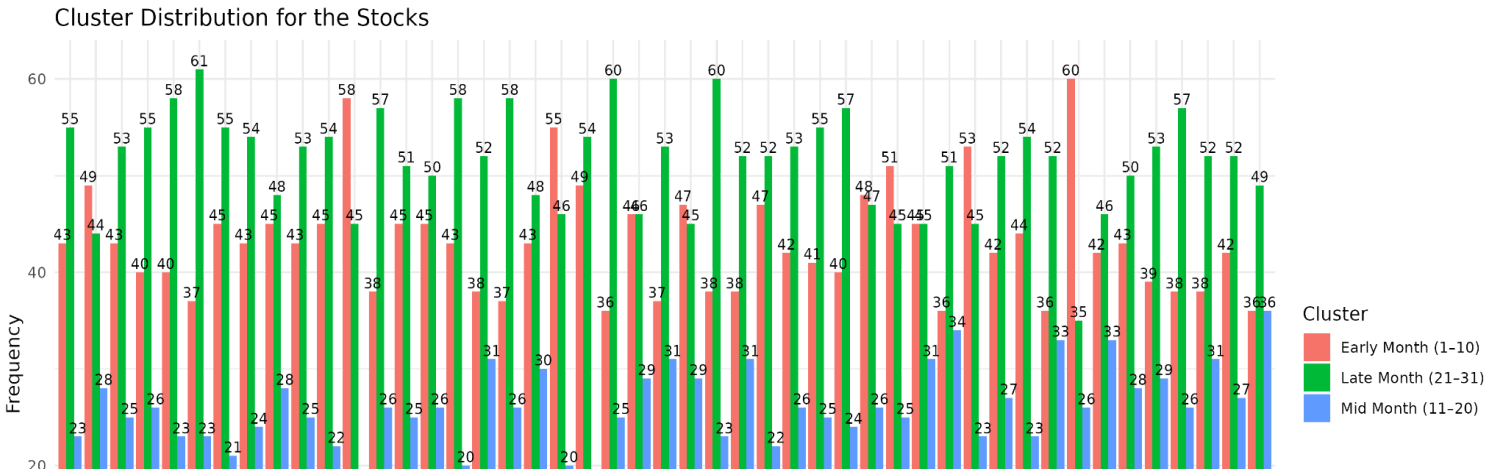

Figure 2

Figure 3

Results

Distribution of Monthly Highs:

Across 5,781 monthly observations for NIFTY 50 stocks from 2015 to 2025, the timing of monthly price highs showed a non-uniform pattern:

- Early Month (1–10): 2,066 highs (35.7%)
- Mid Month (11–20): 1,248 highs (21.6%)
- Late Month (21–31): 2,467 highs (42.7%)

The late-month cluster accounts for almost half of all monthly highs, indicating a persistent concentration of price peaks toward the end of the trading month.

Chi-Squared Goodness-of-Fit Test:

- $\chi^2 = 400.6$, df = 2, $p < 2.2 \times 10^{-16}$.
- Interpretation: The hypothesis of uniform distribution across calendar clusters is decisively rejected. The statistical evidence confirms that monthly highs are not evenly spread and that late-month highs dominate across the sample period.

Sectoral Independence Analysis:

- $\chi^2 = 33.13$, df = 34, $p = 0.51$
- Interpretation: No significant association was found between sectors and the timing of monthly highs. This suggests that the late-month concentration is not sector-specific but a broad market phenomenon, likely reflecting systemic behavioral or institutional trading patterns.

Implications:

The findings highlight a structural calendar effect in the Indian equity market, where price highs tend to cluster at the month's end regardless of sector. This pattern may be linked to portfolio rebalancing, mutual fund window-dressing, or investor sentiment cycles, warranting further investigation into its behavioral and practical trading implications.

Discussion

The observed concentration of monthly highs in the late-month cluster across NIFTY 50 stocks suggests the presence of systematic calendar effects in Indian equities. Several explanations may account for this pattern. First, end-of-month portfolio rebalancing by institutional investors often results in increased trading volumes and price pressures, contributing to upward momentum near month-end. Second, behavioral trading patterns, such as investor optimism surrounding corporate reporting dates or settlement cycles, may also drive this effect. Third, structural features like liquidity and settlement cycles can generate demand-supply imbalances, influencing price peaks at predictable times.

This finding aligns with earlier studies documenting turn-of-the-month effects in global markets, where returns around month-ends systematically outperform other periods. However, unlike the classical turn-of-the-month anomaly focused on average returns, our study identifies a timing bias in monthly highs, adding a new dimension to the understanding of calendar effects.

The absence of significant differences across sectors suggests that the phenomenon is broad-based rather than sector-specific, strengthening the argument for a structural or behavioral cause rather than industry-driven factors.

Limitations of this study include its focus on large-cap NIFTY 50 stocks, which may not fully represent the broader Indian equity market. Extending the analysis to mid- and small-cap stocks or incorporating cross-market comparisons would further validate the robustness and universality of these findings. Additionally, investigating the interplay between institutional trading behavior and market microstructure could yield deeper insights into the mechanisms behind this late-month clustering.

Conclusion

This study provides clear evidence of a late-month bias in monthly price highs across NIFTY 50 stocks over the last decade. The phenomenon is statistically significant and consistent across all sectors, indicating that it reflects a market-wide behavioral or structural tendency rather than industry-specific dynamics. By demonstrating that monthly highs are not evenly distributed throughout the month, this research adds to the literature on calendar-related anomalies and within-month seasonality in equity markets.

The findings have practical implications for portfolio managers, traders, and behavioral finance researchers. Understanding when price peaks are most likely to occur may inform portfolio timing, risk management, and trading strategies. Future research could extend this analysis to mid- and small-cap stocks, incorporate international comparisons, and explore the institutional and behavioral drivers underlying this bias.

References

- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Lakonishok, J., & Smidt, S. (1988). Are seasonal anomalies real? A ninety-year perspective. *Review of Financial Studies*, 1(4), 403–425.
- Ariel, R. A. (1987). A monthly effect in stock returns. *Journal of Financial Economics*, 18(1), 161–174.
- Keim, D. B. (1983). Size-related anomalies and stock return seasonality: Further empirical evidence. *Journal of Financial Economics*, 12(1), 13–32.